

PANKAJ SHARMA

RESEARCH CHEMIST | CHEMISTRY EDUCATOR | ENVIRONMENTAL AND CHEMISTRY
LAB ANALYST | APPLIED NANO CHEMIST

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📍 New Suraksha Vihar, Rohta Gwalior Road, Agra, India

SUMMARY

With a strong foundation in chemistry and hands-on research experience, I am interested in exploring interdisciplinary research across chemistry, materials science, environmental science, nanotechnology, and related fields. My experience includes laboratory experimentation, analytical characterization, environmental monitoring, nanomaterial synthesis, and scientific data interpretation. I am particularly interested in learning new methodologies, working with diverse research groups, and developing expertise across emerging areas of science. I seek research opportunities that encourage continuous learning, interdisciplinary collaboration, and the application of chemistry to solve scientific and real-world problems.

EXPERIENCE

Project Associate (PAT-1) 06/2026 - Present
Indian Institute of Technology, Ropar Punjab

- Work with **Punjab Excise and Taxation Department** and conducted structural and technical audits of distilleries, breweries, bottling plants, and ethanol manufacturing units under the Punjab Excise project.
- Assessed plant layout, production and storage infrastructure, process equipment, utilities, safety systems, fire protection, and environmental management practices for compliance with applicable requirements.
- Prepared technical audit observations and recommendations, identifying infrastructure gaps, operational risks, maintenance requirements, and opportunities for improved safety and regulatory compliance.

Project Associate (PAT-1) 06/2026 – Present
Indian Institute of Technology, Ropar Punjab

- Worked on the green synthesis of carbon quantum dots using low-cost precursors and microwave-assisted methods.
- Focused on controlling particle size, yield, and surface functionalization to enhance fluorescence and selective binding of heavy metals in wastewater.
- Explored applications of CQDs in environmental sensing, pollutant detection, and catalytic degradation of industrial contaminants.

Research Project Assistant 10/2025 – 06/2026
Indian Institute of Technology, Ropar Punjab

- Worked with the **Punjab Pollution Control Board (PPCB)** on water quality monitoring and assessment of the Satluj River.
- Conducted field sampling and laboratory analysis to evaluate overall water quality and pollution levels.
- Gained experience in pollution source identification, environmental data interpretation, and preparation of technical reports in accordance with regulatory standards.

PGT Chemistry Educator and Assessment Coordinator 04/2025 – 10/2025
Kunal Public School Agra

Chemistry Educator and Lab Incharge 02/2022 – 04/2025
S. A. S Educational Institute (Part-time) Agra

Science Teacher and Lab Incharge 02/2022 – 04/2024
R R Inter College (Part-time) Agra

EDUCATION

MASTER OF SCIENCE (Zoology) 2026 – Present
Dr. Bhimrao Ambedkar University Agra

MASTER OF SCIENCE (Chemistry) 2024 – 2026
Dr. Bhimrao Ambedkar University Agra

BACHELOR OF SCIENCE 2021 – 2024
Dr. Bhimrao Ambedkar University Agra

RESEARCH INTERESTS

Nanomaterials
Wastewater Treatment
Fluorescence Sensing
Green Chemistry
Biosensors
Bio-Nano Interface
Nanobubbles and ROS

Carbon Quantum Dots (CQDs)
Surface Functionalization
Catalysis
Sustainable Materials
Analytical Chemistry
Nanobiotechnology

Environmental Nanotechnology
Heavy Metal Detection
Photocatalysis
Environmental Monitoring
Environmental Chemistry
Industrial Pollution Remediation

TECHNICAL & RESEARCH SKILLS

Laboratory & Research: Environmental monitoring, water and wastewater analysis, sample preparation, laboratory experimentation, analytical testing, data interpretation, laboratory safety, and scientific documentation.

Instrumentation: UV-Vis Spectrophotometer, AAS, ICP-OES, flame photometer, DO meter, COD digester, BOD incubator, turbidimeter, conductivity meter, TDS meter, multiparameter meter, centrifuge, autoclave, muffle furnace, filtration and membrane filtration units, digestion unit, micropipette, analytical balance, hot air oven, laminar hood, fume hood, and 4-channel air sampler.

Analytical Parameters: pH, temperature, TDS, TSS, DO, BOD, COD, total and fecal coliform, bioassay, oil & grease, sulphide, phosphorus species, nitrogen species, SAR, RSC, chromium, MLSS, MLVSS, SV₃₀, SVI, fluoride, sulphate, chloride, phenolic compounds, alkalinity, colour, and heavy metals.

Data & Technology: Data interpretation, educational technology integration, scientific software, research documentation, and professional development.

COMPUTATIONAL & VIBE CODING

AI-Assisted Development: Vibe coding, AI-assisted programming, rapid prototyping, software development, workflow automation, and development of research-oriented applications.

Technologies: Python, HTML, CSS, SQL, XML, LATEX, JavaScript, SQLGit, GitHub, VS Code, and AI-assisted development tools.

GitHub Projects:

- **CHEMLAB-PRO** – Laboratory management and research workflow application.
- **Spectrocali** – Spectroscopic calibration and analytical data tool.
- **SmartLIMS** – Laboratory Information Management System.
- **AGRITECH** – Agricultural and experimental data management application.
- **DOI-EXTRACTER** – Research literature and DOI extraction tool.

GitHub: github.com/punitsharma16042003

SOFTWARES & TOOLS

Avogadro
Zotero
Power BI
Microsoft PowerPoint
Adobe Photoshop
QGIS
ChemSketch
DB Browser
Protein Data Bank (PDB)
Fiji
OBS
VS Code
SciDAVis
WRPlot View

GaussView
Origin
Microsoft Excel
Adobe InDesign
GIMP
Fusion 360
Draw.io
PubChem
Overleaf
EPA PMF
Da Vinci
Orange Data Miner
KiCAD
SpectroGryph

BioRender
Google Earth Pro
Microsoft Word
Canva
ArcGIS
ChemDraw
Affinity Publisher
PubMed
Mendeley
Obsidian
TeXwork
Chimera
FreeCAD
R Studio

CERTIFICATIONS & TRAINING

Certificate

Advance Organic Chemistry
Principles of Chemical Science
Satellite Data Analysis for Weather and Climate

Issuing Organization

UCI Open
MIT OpenCourseWare
Indian Institute of Tropical Meteorology
(IITM), Pune

Training

Sophisticated Instrument Operator Training Program
(ED-XRF, XRD, BET, HPLC, SEM, ICP-MS, AAS, FTIR, GC)
Trace Elements Analysis in Water and Solid Samples
(IR-MS liquid and Solid Analyser, ICP-MS, GC, WD-XRF)

Training Organization

IIT Jhodpur

IIT Kanpur

LANGUAGES

English	Proficient
Hindi	Native
German	Beginner

ADDITIONAL COMPETENCIES

Photography	Videography	Photo Editing	Video Editing	Graphic Designing
Technological Integration		Data Management	Report Writting	Knowledge of AI Tools
Knowledge of Scientific Research				